Chapter 12 — Nuclear and Particle Physics

Board-style MCQs · PECTAA Physics 11 · Answer key at end

Practice MCQs

1. The nucleus contains:

(a) electrons and protons **(b)** protons and neutrons **(c)** only protons **(d)** only neutrons

2. Mass number A equals:

(a) protons + neutrons **(b)** electrons + protons **(c)** protons only **(d)** neutrons only

3. Atomic number Z equals:

(a) number of nucleons **(b)** number of electrons only **(c)** number of neutrons **(d)** number of protons

4. One atomic mass unit is about:

(a) 0.511 MeV **(b)** 125 GeV **(c)** 931 MeV **(d)** 1.02 MeV

5. The strongest fundamental force is:

(a) gravity **(b)** electromagnetic **(c)** strong nuclear **(d)** weak

6. The weakest fundamental force is:

(a) weak **(b)** electromagnetic **(c)** gravity **(d)** strong

7. Beta decay is caused by:

(a) weak force **(b)** electromagnetic only **(c)** strong force **(d)** gravity

8. The force between quarks is:

(a) electromagnetic only **(b)** strong force **(c)** weak force **(d)** gravity

9. An antiparticle has:

(a) zero mass **(b)** same charge, opposite mass **(c)** same mass, opposite charge **(d)** double mass

10. The antiparticle of electron is:

(a) proton **(b)** neutron **(c)** positron **(d)** neutrino

11. Pair production requires at least:

(a) 0.51 MeV **(b)** 1.02 MeV **(c)** 931 MeV **(d)** 125 GeV

12. Pair production needs a nucleus to conserve:

(a) spin only **(b)** charge only **(c)** momentum **(d)** mass only

13. Electron–positron annihilation produces:

(a) one photon **(b)** two gamma photons **(c)** a neutron **(d)** a proton

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14. Each annihilation photon has energy:

(a) 1.02 MeV **(b)** 931 MeV **(c)** 0.51 MeV **(d)** 125 GeV

15. In alpha decay, mass number decreases by:

(a) 4 **(b)** 0 **(c)** 1 **(d)** 2

16. In alpha decay, atomic number decreases by:

(a) 4 **(b)** 0 **(c)** 2 **(d)** 1

17. In beta-minus decay, atomic number:

(a) unchanged **(b)** decreases by 1 **(c)** increases by 1 **(d)** increases by 2

18. In beta-minus decay, mass number:

(a) decreases by 4 **(b)** increases by 1 **(c)** decreases by 1 **(d)** unchanged

19. In beta-plus decay, atomic number:

(a) decreases by 1 **(b)** decreases by 2 **(c)** unchanged **(d)** increases by 1

20. Gamma decay changes:

(a) neither A nor Z **(b)** both A and Z **(c)** A only **(d)** Z only

21. Alpha particles have:

(a) variable energy always **(b)** discrete energies **(c)** zero energy **(d)** continuous energies

22. Beta particles have:

(a) fixed energy always **(b)** zero energy **(c)** continuous energies **(d)** discrete energies only

23. Beta particles come from:

(a) electron shells **(b)** proton surface **(c)** nucleus at decay **(d)** outside atom

24. In beta-minus decay, emitted with electron is:

(a) neutrino **(b)** alpha **(c)** antineutrino **(d)** positron

25. In beta-plus decay, emitted with positron is:

(a) electron **(b)** antineutrino **(c)** neutrino **(d)** alpha

26. Leptons include:

(a) proton and neutron **(b)** electron and neutrino **(c)** pion and kaon **(d)** quarks only

27. Hadrons are made of:

(a) photons only **(b)** leptons only **(c)** gluons only **(d)** quarks

28. Baryons consist of:

(a) four quarks **(b)** three quarks **(c)** two quarks **(d)** quark and antiquark

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29. Mesons consist of:

(a) quark and antiquark **(b)** two quarks **(c)** three quarks **(d)** four quarks

30. Proton quark composition is:

(a) ddd **(b)** uuu **(c)** udd **(d)** uud

31. Neutron quark composition is:

(a) ddd **(b)** uud **(c)** udd **(d)** uuu

32. Number of quark flavours is:

(a) 8 **(b)** 6 **(c)** 4 **(d)** 3

33. In beta-minus decay at quark level:

(a) $d \rightarrow u$ **(b)** $u \rightarrow d$ **(c)** $s \rightarrow c$ **(d)** $t \rightarrow b$

34. Exchange particle for electromagnetic force is:

(a) photon **(b)** graviton **(c)** gluon **(d)** W boson

35. Exchange particle for strong force is:

(a) W boson **(b)** graviton **(c)** photon **(d)** gluon

36. Exchange particles for weak force are:

(a) gluons **(b)** photons **(c)** W and Z bosons **(d)** gravitons

37. The Higgs boson was discovered in:

(a) 2015 **(b)** 2017 **(c)** 1916 **(d)** 2012

38. Higgs boson mass is about:

(a) 125 GeV/c^2 **(b)** 931 MeV/c^2 **(c)** 0.511 MeV/c^2 **(d)** 1.02 MeV/c^2

39. The Higgs field gives:

(a) charge to particles **(b)** colour to particles **(c)** mass to particles **(d)** spin to particles

40. Most observable matter in universe is:

(a) solid **(b)** liquid **(c)** gas only **(d)** plasma

41. Matter–antimatter asymmetry is due to imbalance of:

(a) Higgs only **(b)** baryons **(c)** photons only **(d)** gluons only

42. Which is a fundamental particle?

(a) proton **(b)** electron **(c)** pion **(d)** neutron

43. Which force has negligible effect between elementary particles?

(a) strong **(b)** weak **(c)** gravitational **(d)** electromagnetic

44. Particles produced by strong interaction are:

(a) gravitons only **(b)** leptons only **(c)** photons only **(d)** hadrons

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45. Strong nuclear force exists between:

(a) p-p, n-n and p-n **(b)** only p-p **(c)** only p-n **(d)** only n-n

46. When neutron changes to proton, we observe:

(a) alpha decay **(b)** gamma decay **(c)** β^- decay **(d)** β^+ decay

47. A change in Z but not A occurs in:

(a) alpha decay **(b)** gamma only **(c)** fission only **(d)** beta decay

48. Positron was discovered by:

(a) Anderson **(b)** Einstein **(c)** Dirac **(d)** Abdus Salam

49. Dirac predicted:

(a) positron **(b)** neutron **(c)** quark **(d)** Higgs

50. The number of quarks in a neutron is:

(a) 2 **(b)** 5 **(c)** 3 **(d)** 4

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Answer Key with Brief Reasons

1. Answer: (b) protons and neutrons — Nucleons are protons and neutrons.

2. Answer: (a) protons + neutrons — A = Z + N.

3. Answer: (d) number of protons — Z fixes element.

4. Answer: (c) 931 MeV — 1 u c^2 ≈ 931 MeV.

5. Answer: (c) strong nuclear — Relative strength 1.

6. Answer: (c) gravity — Relative strength ~10^{-38}.

7. Answer: (a) weak force — Weak interaction changes quarks.

8. Answer: (b) strong force — QCD binds quarks.

9. Answer: (c) same mass, opposite charge — Antiparticle definition.

10. Answer: (c) positron — Positron is anti-electron.

11. Answer: (b) 1.02 MeV — Electron + positron rest energy.

12. Answer: (c) momentum — Nucleus absorbs recoil.

13. Answer: (b) two gamma photons — Two photons conserve momentum.

14. Answer: (c) 0.51 MeV — Half total rest energy.

15. Answer: (a) 4 — Alpha particle is helium nucleus.

16. Answer: (c) 2 — Two protons lost.

17. Answer: (c) increases by 1 — Neutron becomes proton.

18. Answer: (d) unchanged — Nucleon count stays same.

19. Answer: (a) decreases by 1 — Proton becomes neutron.

20. Answer: (a) neither A nor Z — Only energy state changes.

21. Answer: (b) discrete energies — Two-body decay gives fixed energies.

22. Answer: (c) continuous energies — Energy shared with neutrino.

23. Answer: (c) nucleus at decay — Created in decay.

24. Answer: (c) antineutrino — Antineutrino accompanies β^-.

25. Answer: (c) neutrino — Neutrino accompanies β^+.

26. Answer: (b) electron and neutrino — Leptons do not feel strong force.

27. Answer: (d) quarks — Hadrons feel strong force.

28. Answer: (b) three quarks — Protons and neutrons are baryons.

29. Answer: (a) quark and antiquark — Mesons are q $\bar{q}$.

30. Answer: (d) uud — Two up, one down.

31. Answer: (c) udd — One up, two down.

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32. Answer: (b) 6 — up, down, strange, charm, top, bottom.

33. Answer: (a) d → u — Down becomes up.

34. Answer: (a) photon — Photon carries EM force.

35. Answer: (d) gluon — Gluons carry colour force.

36. Answer: (c) W and Z bosons — Weak bosons.

37. Answer: (d) 2012 — CERN discovery.

38. Answer: (a) 125 GeV/c^2 — About 125 GeV.

39. Answer: (c) mass to particles — Interaction with field gives mass.

40. Answer: (d) plasma — Stars and interstellar matter are plasma.

41. Answer: (b) baryons — Small baryon excess survived.

42. Answer: (b) electron — Electron has no internal structure.

43. Answer: (c) gravitational — Gravity is extremely weak.

44. Answer: (d) hadrons — Strong force makes hadrons.

45. Answer: (a) p-p, n-n and p-n — It acts between nucleons.

46. Answer: (c) β^- decay — β^- emits electron.

47. Answer: (d) beta decay — Beta changes z but not A.

48. Answer: (a) Anderson — Carl Anderson discovered positron.

49. Answer: (a) positron — Dirac theory predicted antiparticle.

50. Answer: (c) 3 — Neutron is udd.

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